

Tutorial 12

1. A negative feedback system as shown in Fig. 1 has an amplifier of gain 100 with $\pm 10\%$ tolerance in the forward path, and an attenuator of value $9/100$ in the feedback path. The overall system gain is approximately:

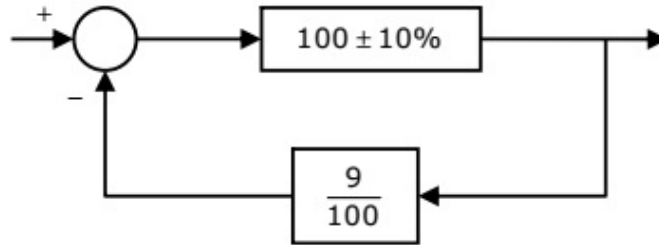


Fig. 1

2. The transfer function $\frac{V_2(s)}{V_1(s)}$ of the circuit shown below is (Refer Fig. 4)

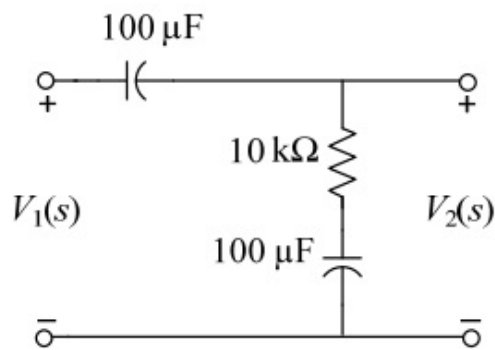


Fig. 4

3. For the block diagram shown in the figure 5, the transfer function $\frac{Y(s)}{R(s)}$ is

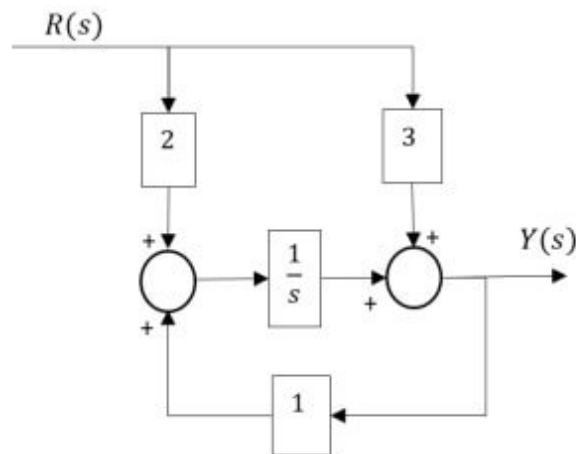
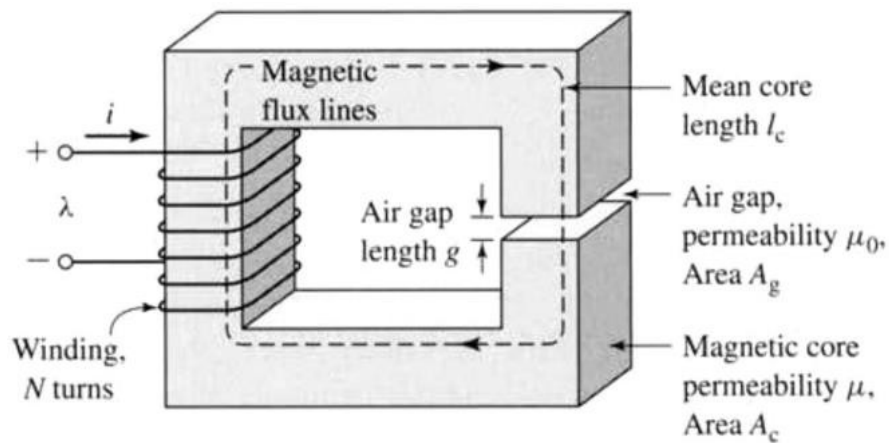
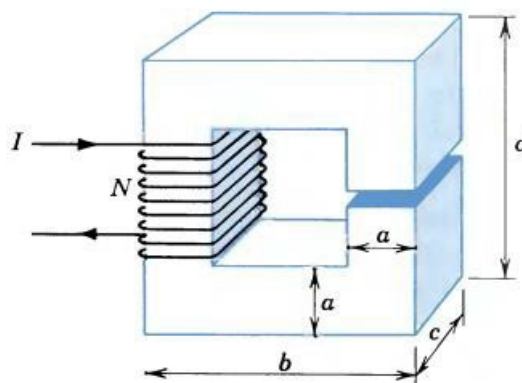


Fig. 5

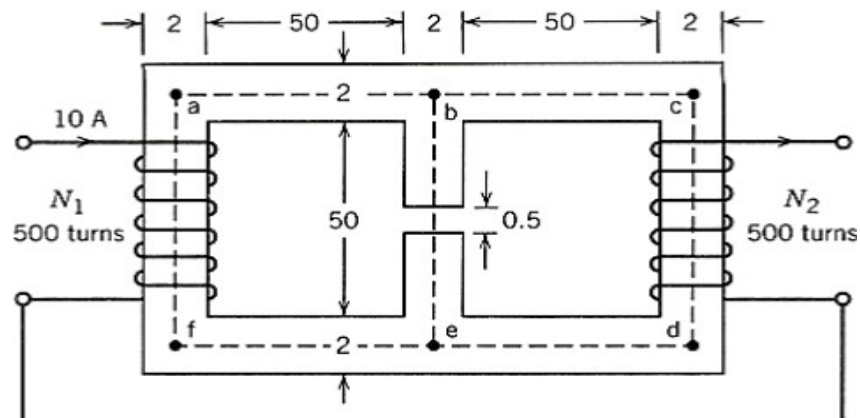
4. Given the magnetic circuit shown below has dimensions $A_c = A_g = 9 \text{ cm}^2$, $g = 0.05 \text{ cm}$, $l_c = 30 \text{ cm}$, $N = 500 \text{ turns}$, $\mu_r = 70,000$ for core material, find (a) the reluctances R_c and R_g , (b) flux and (c) the current i ($B_c = 1.0 \text{ T}$).



5. For the Iron core of fig. given below with $a=5 \text{ cm}$, $b=20 \text{ cm}$, $c=10 \text{ cm}$ and $d=20 \text{ cm}$, $N=200$. Find the current required to establish 6.5 mWb (a) across 5 mm air gap (b) if there is no air gap. Given $\mu_r=4138$



6. In the magnetic circuit shown in the figure, the relative permeability of the ferromagnetic material is 1200. Neglect magnetic leakage and fringing. All dimensions are in cm, and the magnetic material has a square cross-sectional area. Determine the air gap flux, the air gap flux density, and the magnetic field intensity in the air gap.



7. It is desired to establish a flux of 0.5mWb in a toroidal iron core with 400 turns and relative permeability $=1000$. The iron core has cross sectional area of 10^{-3}m^2 and $R=0.04\text{m}$. Find the reluctance and the current required.
8. A 10 kVA, single-phase transformer has a transformation ratio of 0.083 and is supplied from a 2.4 kV supply. Neglecting losses, determine:
 - (a) The full-load secondary current
 - (b) The minimum value of load resistance which can be connected across the secondary winding without the kVA rating being exceeded
 - (c) The primary current
9. An ideal transformer is used inside a DELL laptop charger (20 V, 65 W output). Design this transformer: determine the winding ratio and the Ampere rating of primary side cable connecting the charger to the wall socket supply at your home.